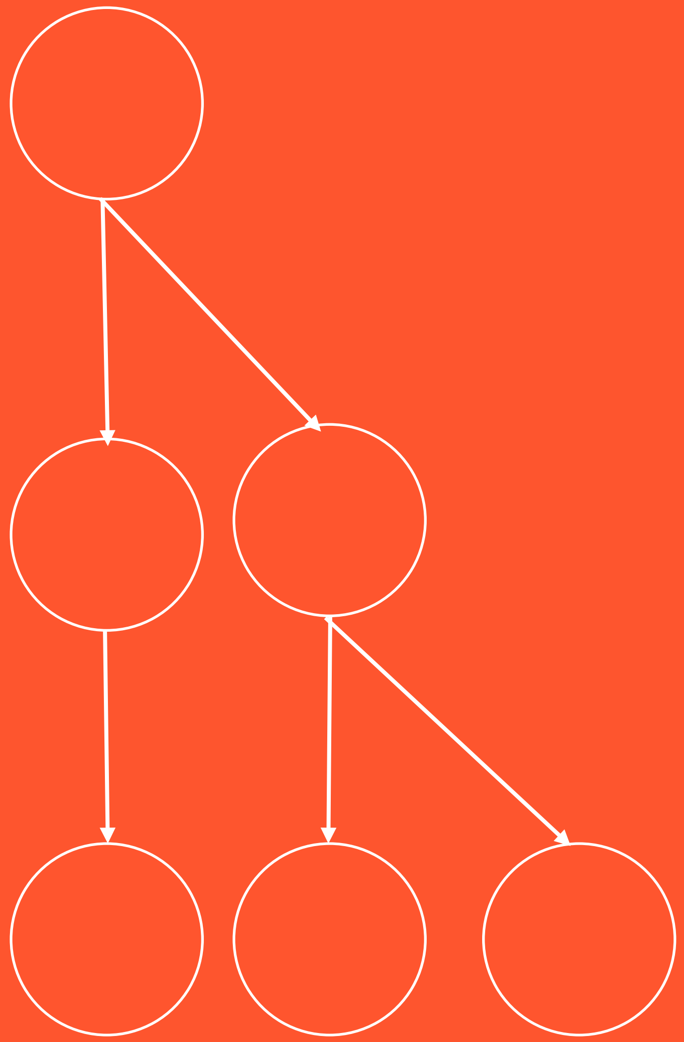


BCOG 100

Memory, Simulation, & Generative Models

Lecture 2: Algorithms for Memory and Simulation

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Algorithms for Memory and Simulation

Computation
What does the system do?

Algorithm
How does the system do it?

Implementation
How is the system realized?

Recap & Overview

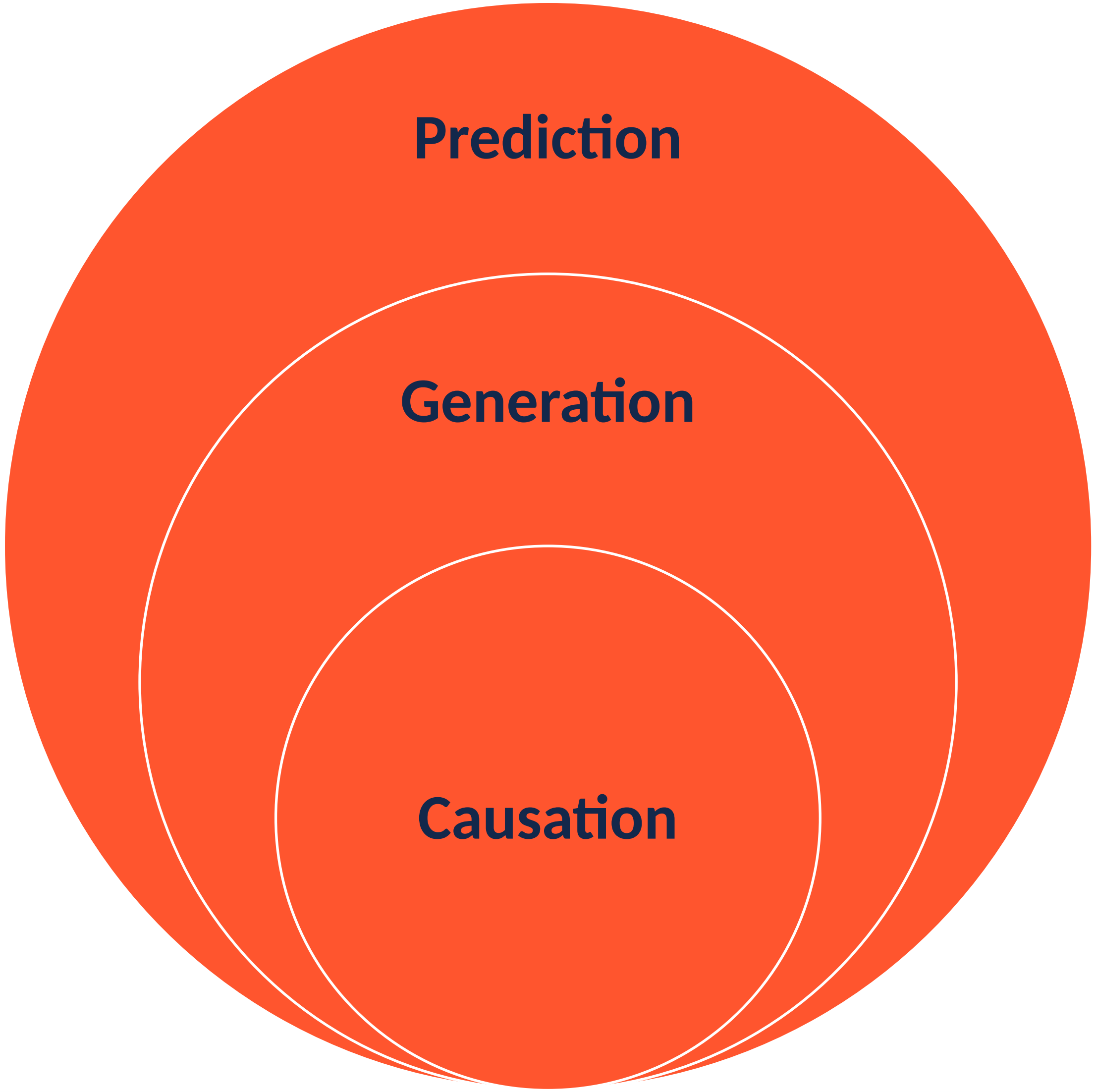
Last Time:
Why simulation evolved.

Today:
- *How* it works.
- From prediction → generation → causation: a hierarchy of modeling power.

Algorithms for Memory and Simulation

A Hierarchy of Models

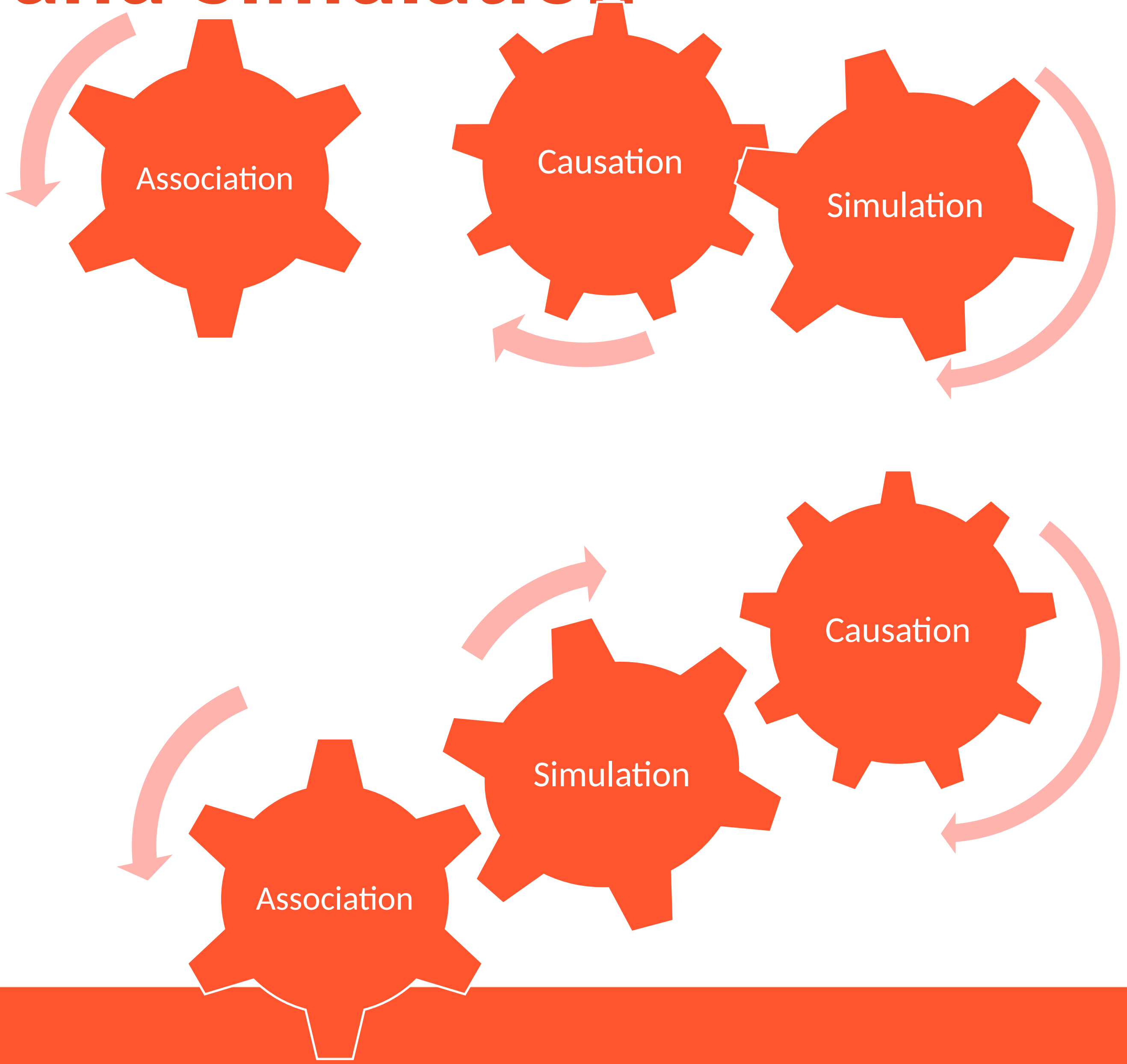
- Prediction: What usually follows.
- Generation: What could follow.
- Causation: What makes it follow.



Algorithms for Memory and Simulation

From Learning by Association to Learning by Inference

- Associative learning links events.
- Inference explains events.
- What is the relationship of generative models to association-based and causal-based knowledge?



Reading Check 9-2

Password: imagine

Enter your answer from Monday. When you're done, raise your hand.

Memorize the list of words that will appear on the right.

In 30 seconds, I am going to ask to you remember as many as you can. Type them into the reading check for today's credit.

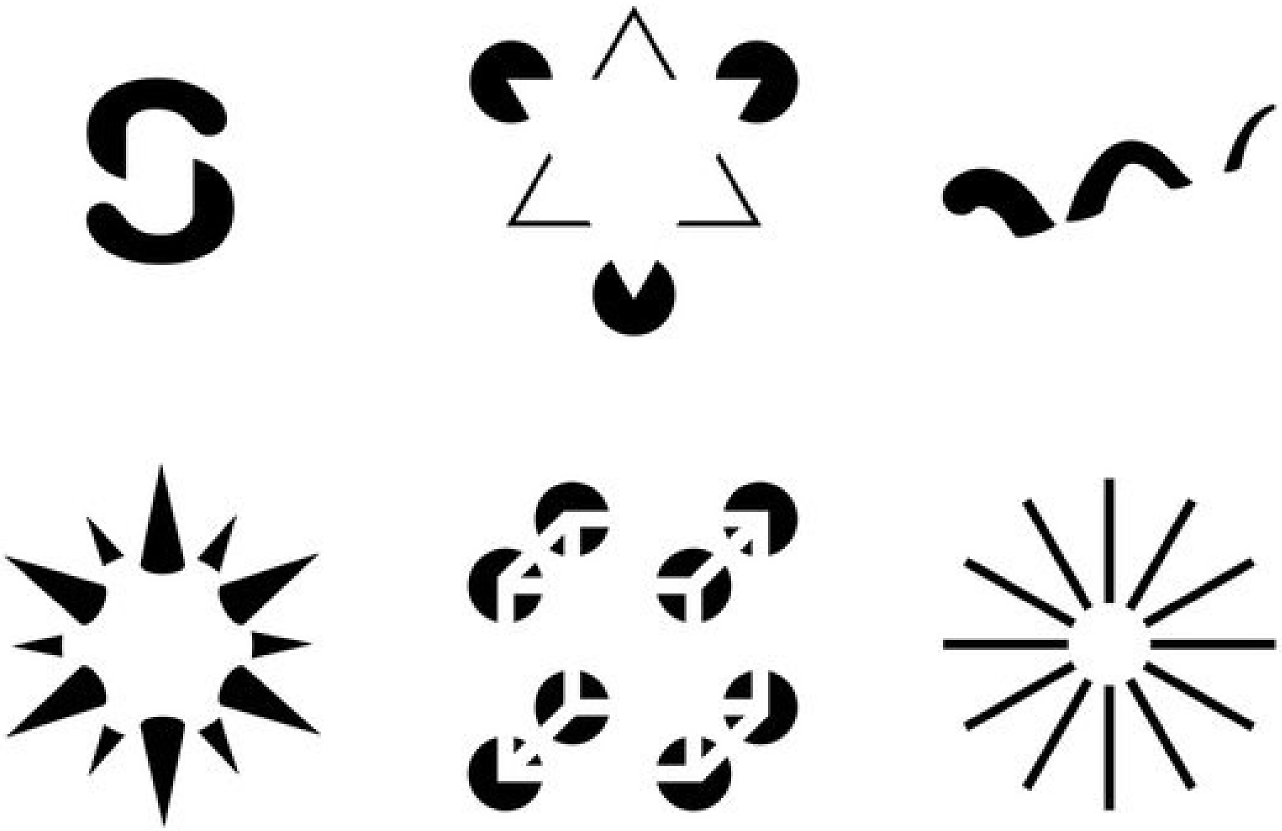
hot
snow
warm
winter
ice
wet
frigid
chilly
heat
weather
freeze
air
shiver
Artic
frost

snow
cold
shoe

Algorithms for Memory and Simulation

Helmholtz and “Perception as Inference”

- We never see the world directly.
- We infer it from incomplete, noisy data.
- Perception is the brain’s best guess.



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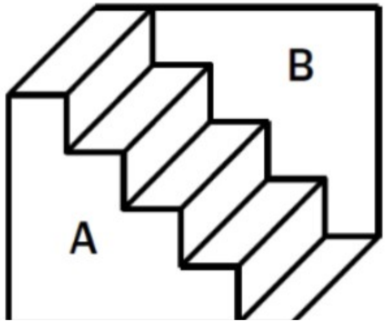
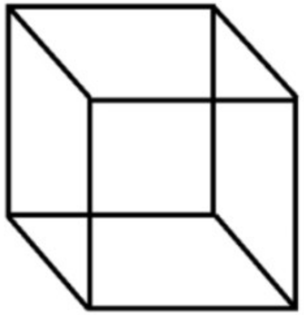
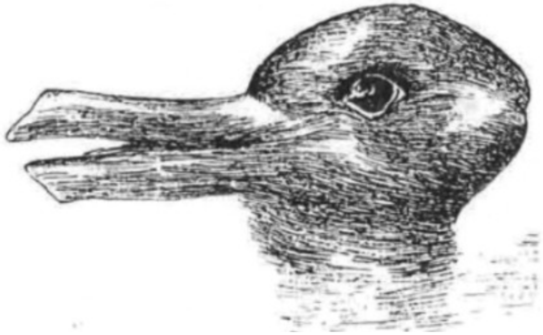
Recognition by Simulation

Perception = prediction tested by sensation.
The brain simulates possible worlds.
The “best-fitting” simulation becomes reality.







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Perceptual Clues

- **Fill in**: The brain invents missing detail.
- **One-at-a-time**: Perception is serial, not simultaneous.
- **Can’t unsee**: Once the model locks in—you can’t unsee it.

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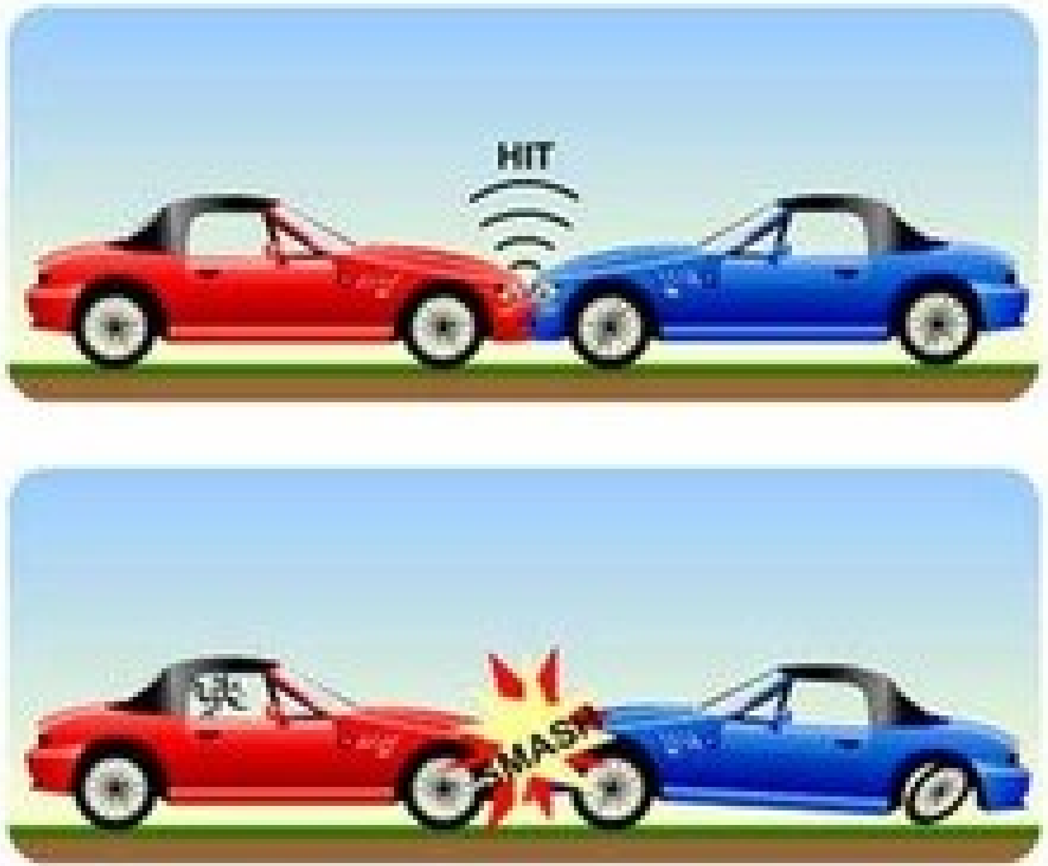


Memory as Reconstruction

- Memory isn't playback—it's reconstruction.
- Each recall is a new simulation.
- Remembering and imagining use the same machinery.

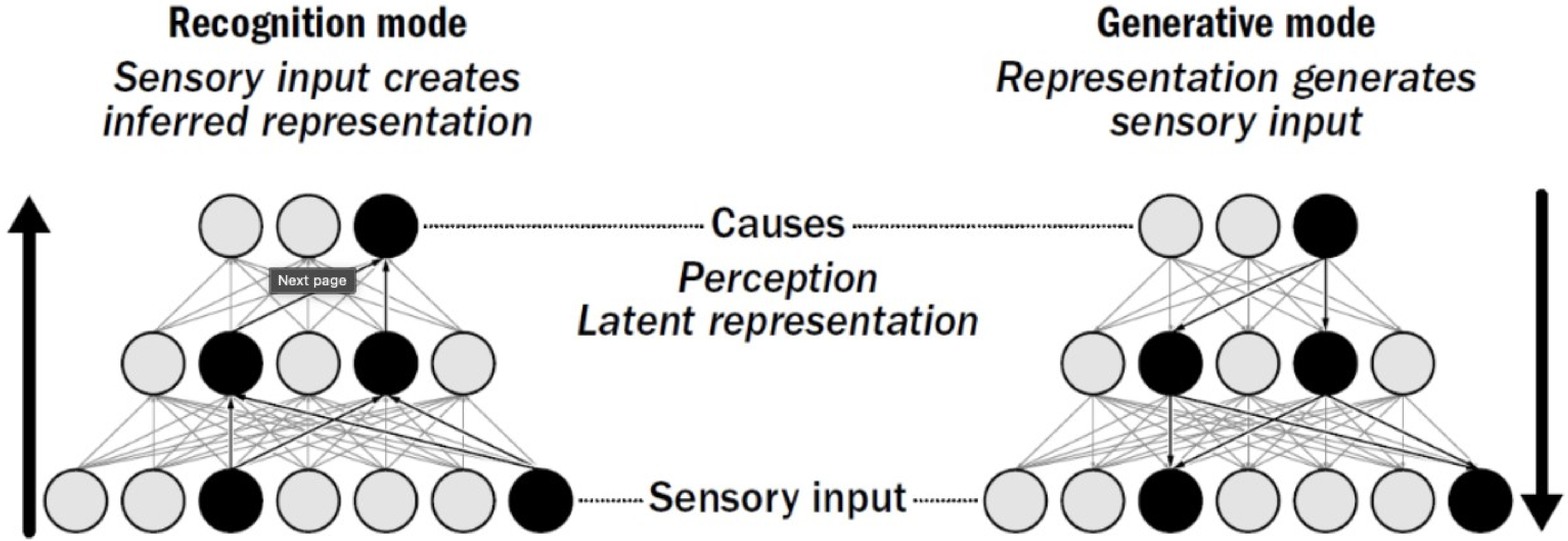
hot
snow
warm
winter
ice
wet
frigid
chilly
heat
weather
freeze
air
shiver
Artic
frost

snow
cold
shoe

Algorithms for Memory and Simulation

The Helmholtz Machine and Early Generative AI

- The same logic powers perception, memory, and learning.
- Predict → compare → correct.
- Early generative networks made this process explicit.



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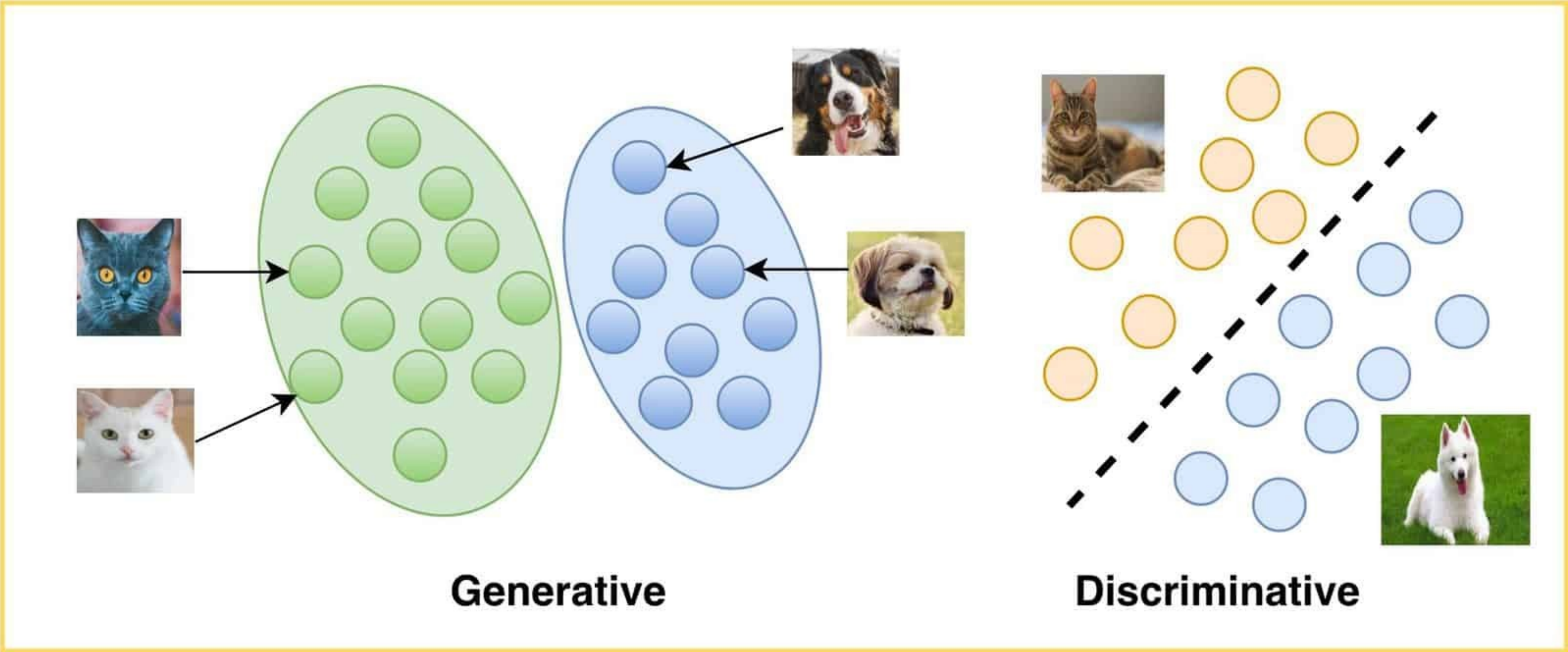
Generative Models
Learn Without Labels

- No teacher needed.
- Prediction itself is the lesson.
- Learning happens by reducing surprise.



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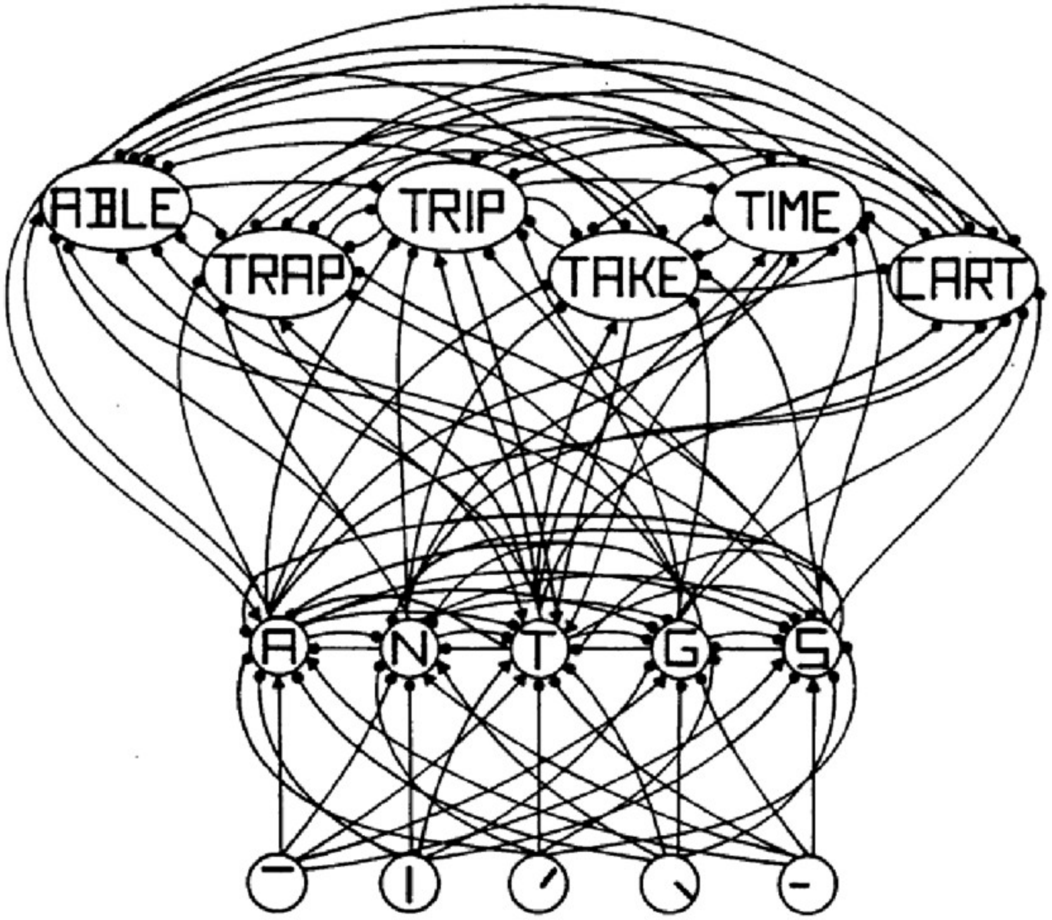

Prediction Error Minimization – The Core Algorithm

- Brains and models learn by reducing surprise.
- Prediction errors are not mistakes—they're information.
- Every perception is an experiment.

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Perception as Constrained Hallucination

- Your brain is always hallucinating.
- Sensation just keeps it in check.
- Prediction creates perception.

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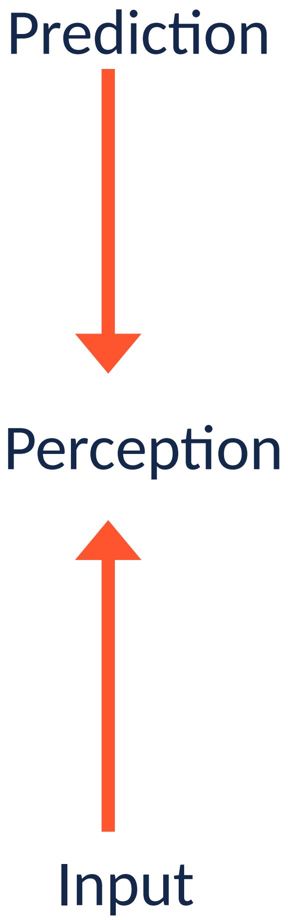

Prediction and Input in Balance

- Perception = prediction + evidence.
- Too much prediction → hallucination.
- Too much input → noise and confusion.
- The mind needs both to stay grounded.

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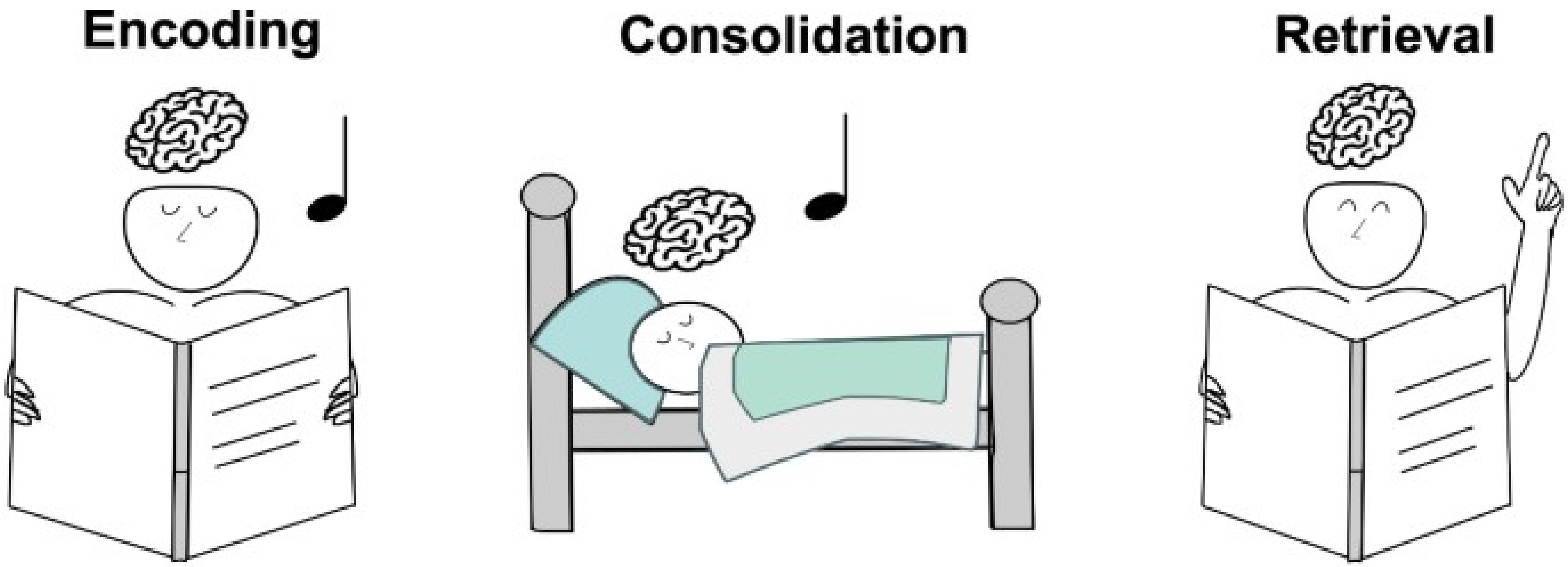

Generative Replay and Memory Integration

- Brains learn by replaying imagined experience.
- Memory consolidation = internal simulation.
- Old and new experiences merge through generative replay.

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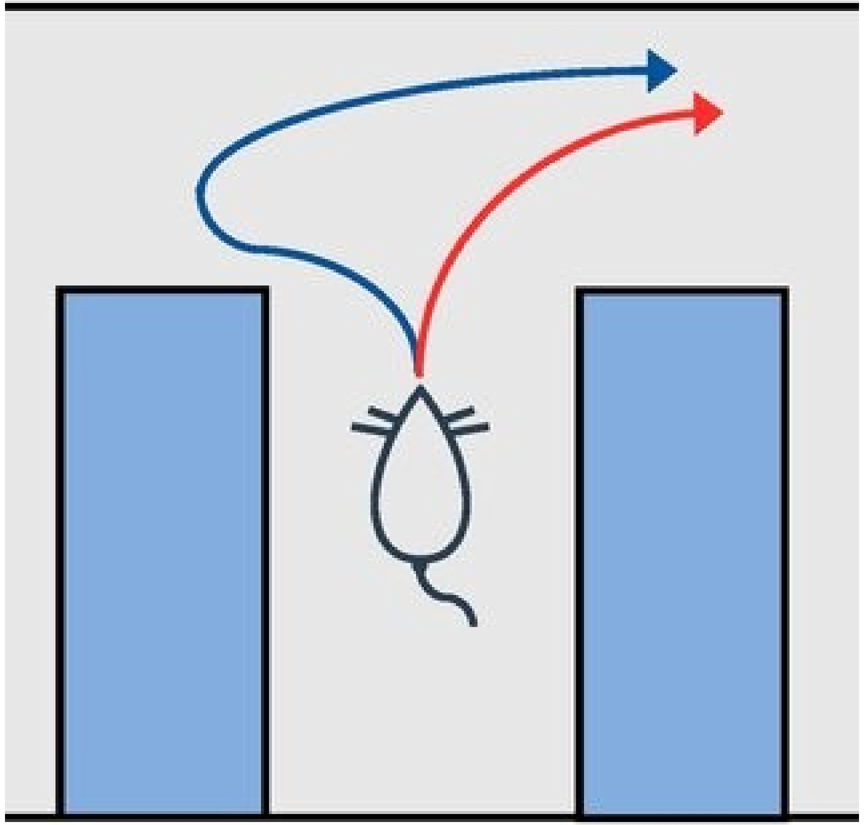

Counterfactual Simulation – Thinking "What If."

- Generative models can imagine alternatives.
- Counterfactuals expand learning beyond experience.
- "What if" is prediction untethered from history.

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From Predictive to Generative – But Not Yet Causal - Models

- Prediction links events.
- Generation simulates possibilities.
- Causation explains *why*—a further leap still.

Model Type	Learns from	Can do	Cannot do
Predictive	Repetition	Expect next event	Imagine new ones
Generative	Simulation	Imagine new outcomes	Explain why they occur
Causal	Intervention	Predict and explain	—

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World Models in AI and Brains

- Brains and AI both build internal simulations.
- A “world model” predicts consequences of actions.
- But prediction ≠ understanding.

Figure 11.9: StyleGAN2 from thispersondoesnotexist.com

Pictures from thispersondoesnotexist.com

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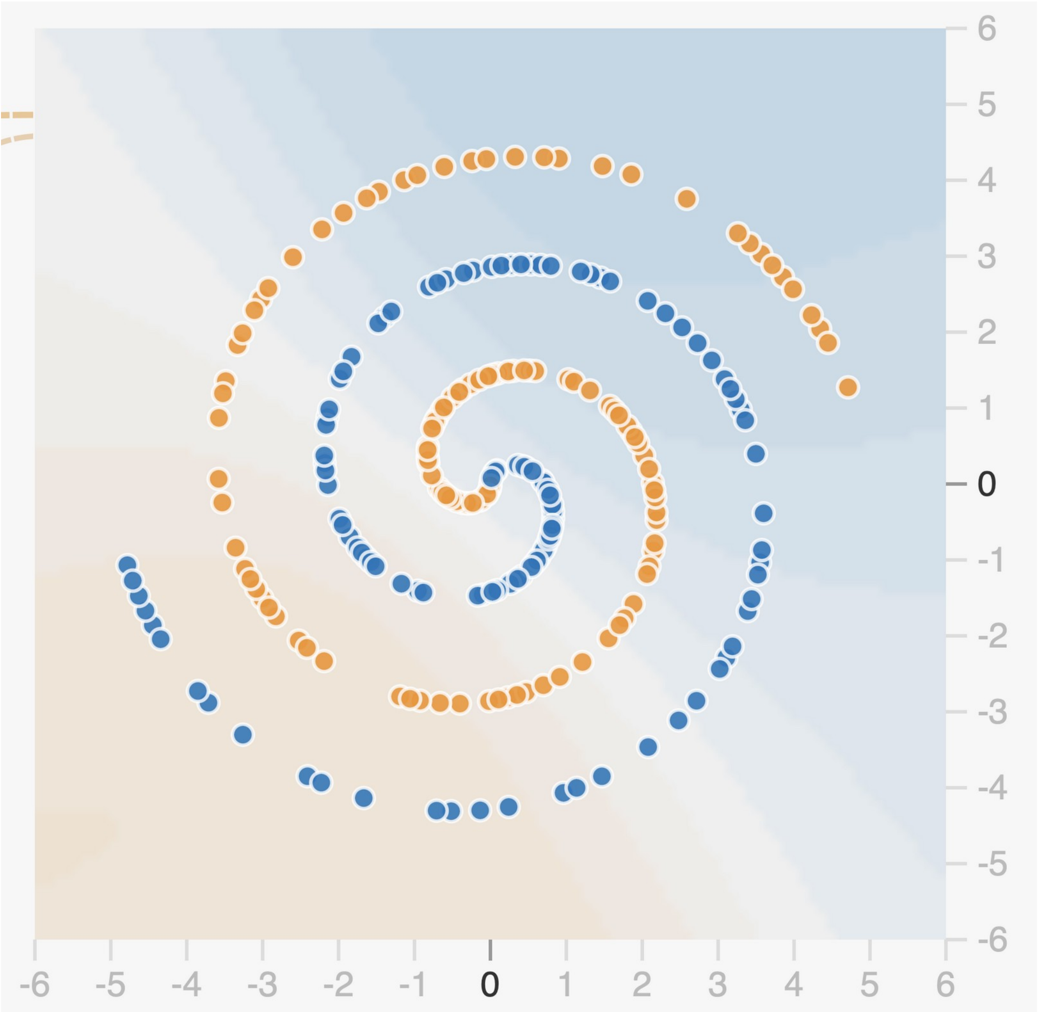
Limits of Generative Models

- Generative models predict patterns, not principles.
- They imitate structure without knowing cause.
- When the world changes, they fail.

Ice Cream Consumed

Hot Weather ?

Number of Drownings



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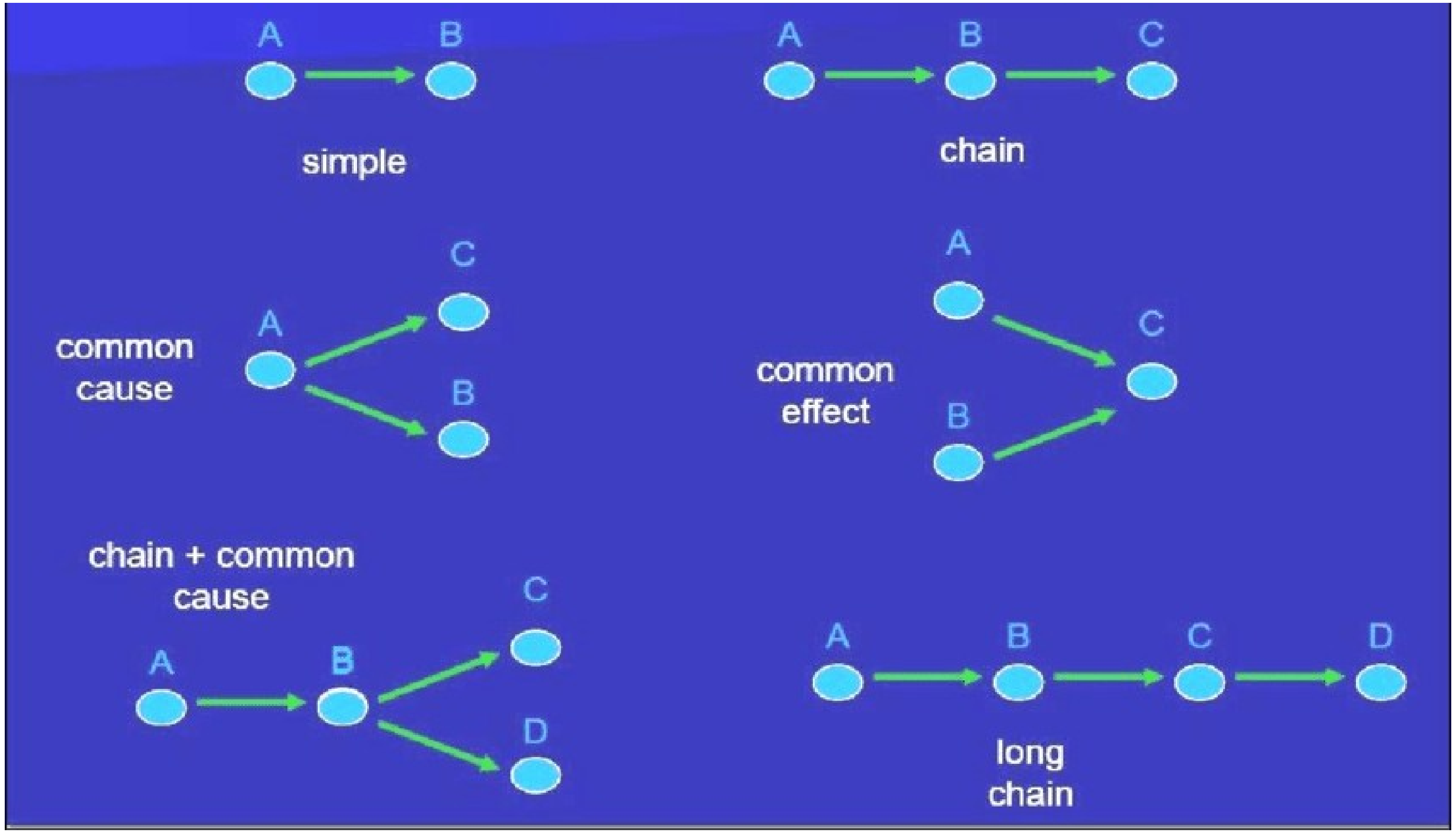
What Makes a Model Causal?

- Causal models represent interventions.
- They can ask: “What if I change this?”
- Understanding = knowing how causes make effects.



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Evidence for Causal Reasoning Across Species

- All vertebrates predict.
- All mammals simulate.
- Few represent true cause and effect.
- Causal reasoning emerges late—and grows with language.

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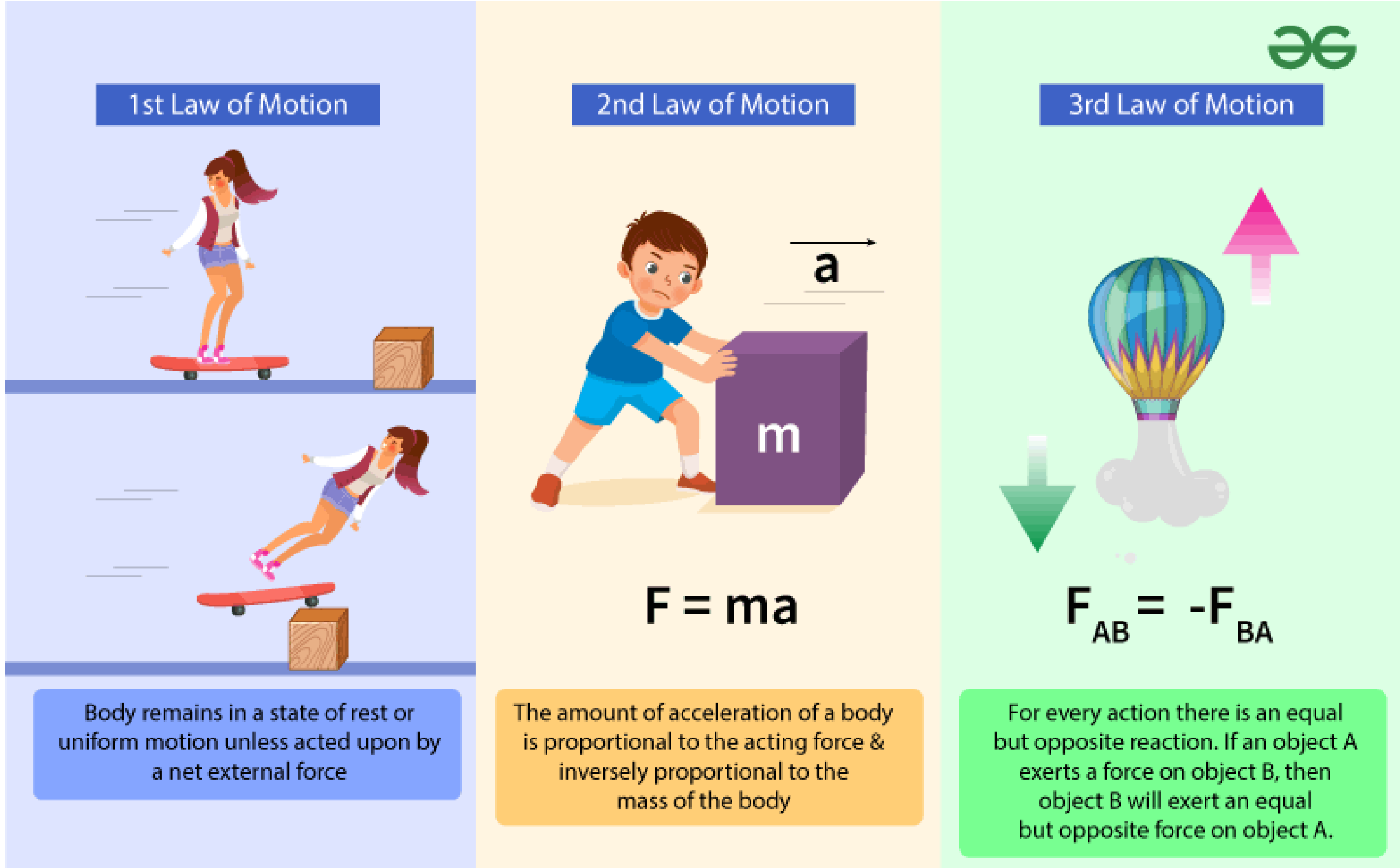
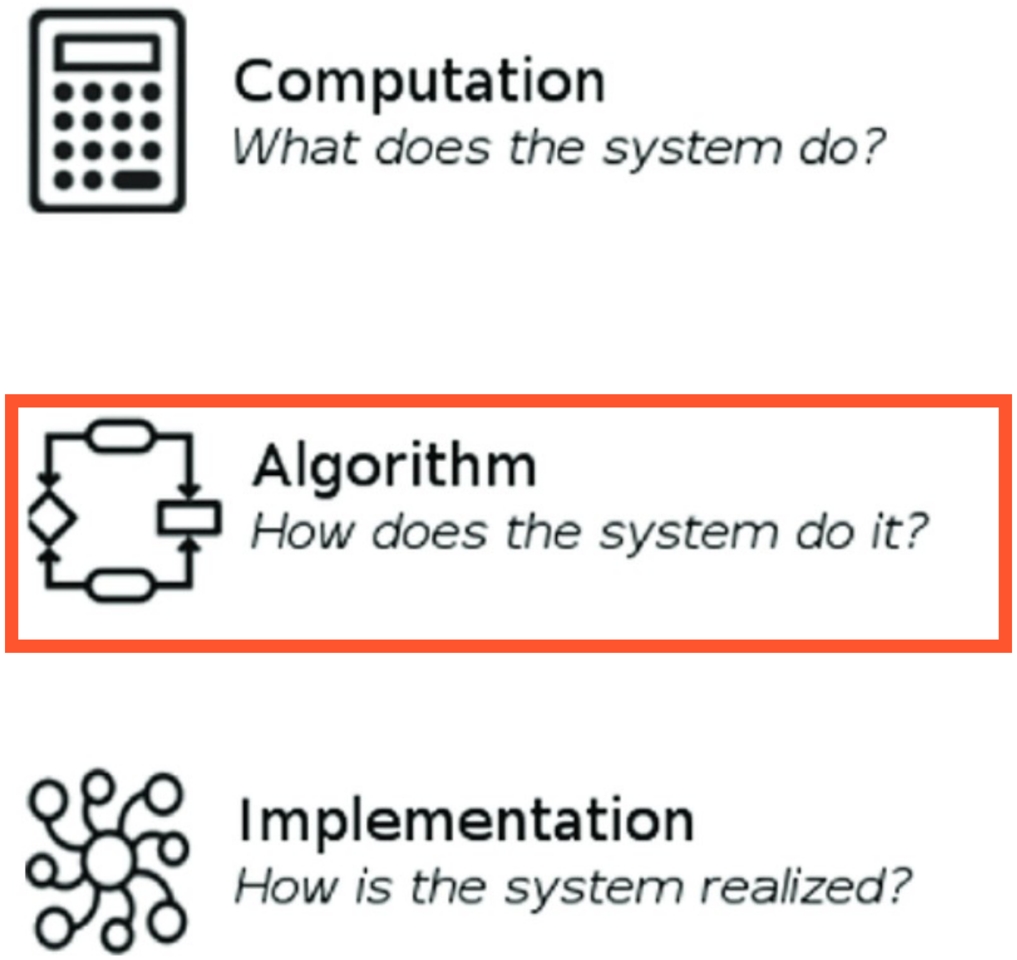
Humans and the Cultural Amplification of Causality

- Language makes causes explicit.
- Culture preserves and refines them.
- Science is collective causal modeling.



The Evolution of the Cortex



Summary

- Prediction → Generation → Causation
- Each level adds representational power.
- Brains predict.
- Minds imagine.
- Cultures explain.

Next Time

- Simulation in the brain and the structure of the neocortex